EDMS 7-400-2

SPECIFICATION

FOR

DISTRIBUTION KIOSKS (GRP)
FOR TRANSFORMERS UP TO 3000 KVA
22-(11- 10.5)/0.4 KV

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Ver.2

EEHC

توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

CONTENTS

1. SCOPE

- This specification describes the general requirements governing the design, manufacturing and testing of the prefabricated high-voltage/low voltage substation with glass fiber reinforced polyester (GRP) housing up to 24 kV type tested according to IEC 62271-202 ed.2.0 standard.
- The prefabricated high-voltage/low voltage substation is commonly called Compact Secondary Substation (CSS) and it can be called Package Substation or Package Distribution Substation.
- The Compact Secondary Substation (CSS) comprises of an enclosure, transformer, Medium Voltage (MV) switchgear or Ring Main Unit (RMU), Low Voltage Switchboard (LVS), medium voltage and low voltage interconnections and auxiliary equipment to supply low voltage energy from a medium voltage system or vice versa
- The CSS enclosure expected operating lifetime is 30 years under the specified environmental, climatic, system and service conditions specified, with minimum maintenance requirements.
- The power rating of the CSS will be defined as per the rating of the installed transformer so different power ratings of the CSS are available.

2. APPLICABLE CODES & STANDARDS

The specifications shall be read in conjunction with the latest version of EDMS specifications

Table (1)

EDMS 2-100	12 KV AIR RING MAIN UNITS
EDMS 2-102	TECHNICAL SPECIFICATIONS FOR 12 KV RING MAIN UNIT (GIS / AIS)
EDMS 2-200	24 KV AIR RING MAIN UNITS
EDMS 2-202	TECHNICAL SPECIFICATIONS FOR 24 KV RING MAIN UNIT (GIS / AIS)
EDMS 02-101	TECHNICAL SPECIFICATION FOR 12 KV AUTOMATED SMART RMU (GIS / AIS)
EDMS 02-103	TECHNICAL SPECIFICATION FOR 12 KV NON-MOTORIZED RING MAIN UNIT (GIS / AIS)
EDMS 02-203	TECHNICAL SPECIFICATION FOR 24 KV NON-MOTORIZED RING MAIN UNIT (GIS / AIS)
EDMS 02-201	TECHNICAL SPECIFICATION FOR 24 KV AUTOMATED SMART RMU (GIS / AIS)
EDMS 8-100	SPECIFICATIONS FOR OIL – IMMERSED DISTRIBUTION TRANSFORMER FROM 25 KVA TO 5000 KVA 10.5- 11/ 0.4 KV
EDMS 8-200	SPECIFICATIONS FOR OIL – IMMERSED DISTRIBUTION TRANSFORMER FROM 25 KVA TO 5000 KVA 22/ 0.4 KV
EDMS 8-101	SPECIFICATIONS FOR DRY TYPE TRANSFORMER FROM 50 KVA TO 5000 KVA 10.5- 11/ 0.4 KV EPOXY SOLID – CAST / CAST RESIN COIL
EDMS 8-202	SPECIFICATIONS FOR DRY TYPE TRANSFORMER FROM 50 KVA TO 5000 KVA 22/ 0.4 KV EPOXY SOLID – CAST / CAST RESIN COIL
EDMS 19-400	TECHNICAL SPECIFICATION FOR HIGH RUPTURE CAPACITY MV CURRENT LIMITING FUSES (HRC) 12 KV & 24 KV

The CSS enclosure, installed electrical and the relevant equipment shall be designed, manufactured and tested according to the latest version of IEC standards as per the below:

Table (2)

IEC 62271-202	High-voltage / low voltage prefabricated substations
IEC/TR 62271-208	Electromagnetic fields generated by CSS
IEC 60529	Degree of protection provided by an enclosure (IP code)
IEC 62262	Degree of protection against external mechanical impacts (IK code)
IEC 60721	Classification of environmental conditions

3. ENVIRONMENTAL AND SYSTEM CONDITIONS

The substation is preferred to be used under field sites which may be exposed to severe weather and/or vandalism conditions for outdoor switchgear and control gear according to IEC 62271-1 (Latest version)

Service conditions

Table (3)

Minimum ambient Temperature	-5°C
Ambient Temperature (max.)	+45 °C (50 °C as option)
Relative humidity	up to 95%
Altitude (above sea level)	up to 1000m

4. DESIGN CRITERIA:

1) The CSS shall be of Glass Reinforced Polyester (GRP) construction. This applies also to the roof, doors and ventilation louvres.
2) CSS shall facilitate the installation/ removal of any equipment (transformer- RMU – LV panel) the arrangement from the doors.
3) CSS shall be provided with lifting holes from bottom so that the whole unit can be lifted and positioned on concrete plinth at site by crane without causing any damage or distortion
4) The CSS and the installed components shall be designed and manufactured to operate properly and withstand:
   - The climatic & environmental conditions stated above.
   - The electrical system parameters stated above.
5) The enclosure should be a robust design. The CSS enclosure shall operate safely in order to provide the needed level of safety and protection from un-authorized access.
6) Construction shall conform to IEC 62271-202:2014. This standard specifies the service conditions,

rated characteristics, general structural requirements and test methods for prefabricated substations.

7) Each compartment shall have its own dedicated access and be segregated by sheet steel compartment walls with min. thickness of 2 mm for 11 KV (2.5 mm for 22 KV) (before painting) to protect an operator from a possible catastrophic failure of electrical apparatus in an adjacent compartment.
8) Each wall panel shall be constructed with a double layer GRP encapsulated an overall thickness of 50 mm and outer layer thickness not less than 4 mm.
9) The design of the CSS shall be based on safety to personnel and equipment during operation and maintenance. It shall be internal arc proof tested type (IAC-AB) to the arc current levels (IAC AB, 20 kA at 1 sec) to provide a high level of personal safety to the general public and operators.
10) Any metal part as standard shall be made of galvanized sheet steel and fasteners shall be aluminum, zinc-coated or of stainless steel in order to provide protection against corrosion.
11) Any fixing between any two GRP parts and plinths shall be made internally.
12) All the auxiliary connections should be executed using copper conductors (450/750) V fire-proof insulated stranded copper conductors of cross-sectional area at least (2.5 mm2) for current circuits, and (1.5 mm2) for voltage circuits.
13) The insulation power frequency withstands voltage for low voltage switchgear, and secondary circuits should be according to the latest IEC
14) Enclosure finishing should assure minimal outer surface deterioration and shall be of a pigmented gel coat.
15) The enclosure shall withstand all mechanical stresses including roof-load, wind-loads, and external mechanical impacts. Covers, doors, and ventilation openings shall be able to withstand mechanical impacts corresponding to a degree of protection IK10.
16) The enclosure shall be designed in such a way that the wall panels design ensures no water ingress between the base frame and the housing.
17) The GRP enclosure material shall be inflammable, such that it shall have one-hour fire retardant construction according to ISO 834-1 or corresponding national standard. GRP material shall be self-extinguishing according to ASTM D635; with flammability classification V-0: UL94 or corresponding national standards.
18) Transformer compartment shall be equipped with the oil collection pit underneath the transformer (in case of using an oil transformer). Oil collection pit is intended to trap, in case of a transformer failure, the insulating fluid flowing out of the transformer, protecting the ambient environment and groundwater. Oil collection pit is a sealed, fully confined, oil- and watertight solution without a supplementary coating.
19) The temperature class of the transformer compartment shall adequate class-10. The manufacturer needs to produce evidence about its compliance to the designated ‘temperature class’ by conducting heat-run test as per the IEC 62271-202:2014 or by using a reliable software which can simulate such performance.
20) CSS roof shall be designed as a separate constructional component which can be lifted off on site when replacement of equipment is needed. Roof fixation shall be easy-to-access and detachable after opening the doors.

21) The roof shall be designed to support an equally distributed load of minimum 2500 N/m^2 without deformation.

22) The roof of the CSS should have a slope (not less than 6 degrees) to prevent the accumulation of rainwater.

23) The CSS should be totally enclosed, and weather-proof since it should be located in the open air.

24) The CSS doors should be provided with suitable gaskets, and should have locks with a master key (as an option according to …. EDC requirements)

25) CSS door's constructive structure shall be the same of the rest of the enclosure. The doors shall be robust to offer safety and protection against the vandalism.

26) Suitable door shall be provided to enable personal to operate, repair, maintain or remove equipment. The size of the doors shall be designed to give maximum opening access with regard to type and size of enclosure.

27) All doors shall comply with internal arc fault testing criteria regarding pressure withstand.

28) All doors shall be provided with stoppers and to be hung by stainless steel T hinges secured by stainless steel bolts, washers and nylon nuts.

29) All doors shall be fitted with heavy duty 2 point locking systems, equipped with handle and covered place for padlock. One MV or LV door shall have a drawing pocket inside.

30) Except earthing copper bus bars, all main medium voltage bus bars should be fully insulated with anti-track heat-shrinkable tubes of voltage (12/20), (18/30) kV, and all low voltage bus bars should be insulated with colored heat-shrinkable tubes (red, yellow, blue, and black) of voltage (0.6/1) KV

31) The degree of protection of the whole substation including doors and ventilation openings shall be at least IP23. Medium Voltage and Low Voltage compartments shall be rated with IP54 as minimum.

32) Interconnections between the MV switchgear and the transformer shall be made of suitable cable with the permitted bending radius. Interconnections between LV and transformer shall be made of copper busbar, or flexible copper strap between the main busbar and bushing of transformer, preventing damage of bushing due to transportation vibration.

33) The cable connecting the fused load-break switch to the transformer should be XLPE insulated with a copper conductor of cross-section area not less than 3x1x35 mm^2, 12/20 KV , 18/30 KV for transformers up to 500 KVA, and 3x1x70 mm^2,12/20 KV, 18/30 KV for transformers more than 500 KVA, all with screen copper tapes [or tape & wires].

34) The cable should be terminated with heat-shrinkable terminations or equivalent 12/20,18/30 KV.

35) Sufficient ventilation shall be installed to meet the installed equipment's environmental requirement.

36) Each load break switch compartment should be equipped with a suitable clamp or any other means for supporting when fixing the cables or end boxes.

37) Suitable inspection windows covered with hard transparent material should be provided as they are necessary.

38)The load-break switch is preferred to be the rotatory type which is less affected by the dust.

39)Each load break switch should have a separate door; each door should have a rubber-sealing gasket.

40) The load break switches should be hand-operated through levers. Their opening, and closing should be quick-acting due to energy stored in the relevant springs.

41)The medium voltage compartment (RMU is 2+1) should have a suitable space for one additional cable load break switch to be mounted in the future if required.

42)RMU could be 3+1 as an option according to… EDC requirement.

43) The operating handle should be inserted in its place when opening or closing any of the switches without dismantling the protective panel.

44) Marking labels should be fitted to indicate the on-off position of the switches.

45)The medium voltage compartment should have sliding button openings (provided with rubber gasket) to permit the entrance of (two or more) ring cables of cross-sectional area up to(3x300)mm^2.

46)The connection terminals of medium voltage, and low voltage switchgear should be suitable for connection to aluminum or copper cables.

47)The low voltage compartment should have sliding button openings for the exit of outgoing feeders' cables.

48)The panel`s doors should be earthed with a mean of copper conductor with cross-section area at least (6 mm^2).

5. TECHNICAL SPECIFICATION FOR M.V. COMPARTMENT

1) RMU should be according to the latest revision of EDMS technical specifications for RMU.
2) The leakage path of the insulators, current transformers, the potential transformers, and the LBS arm should not be less than 2 cm/kV of the nominal operating voltage.
3) Each compartment should be equipped with an indication lamp (LED) for voltage phases (L1, L2, L3), and should be fed from a capacitive divider for cable LBS.
4) The medium voltage compartment should comprise two / (three as an option) cable load break switches 630 A with earthing switches as per attached specification.
5) In the case of 2+1 RMU, the medium voltage compartment should have space to add an additional load break switch in the future.
6) RMU panel (complete with all sides, roof, and bottom) should be put inside at the medium voltage compartment of the CSS.
7) One fused load-break switch without earthing switch for transformer, complete with three high rupture capacity (H.R.C) fuses according to the latest revision of EDMS 19-400.
8) Earth leakage indicator automatic reset working on 220 V (110V if VT available), 50 Hz, complete with Ferranti current transformers which will be mounted around the cable sheath, as per attached

specification. Each earth leakage indicator should be flush mounted.

9) The earthing copper braid of the terminations should be earthed at the side which is connected to the Ring Main Unit.

10) All internal wiring between C.T, and earth leakage indicator, and those for illumination should be enclosed in non-flammable flexible PVC tubes or ducts.

6. HIGH RUPTURE CAPACITY (H.R.C) FUSES

HRC fuse should be according to the latest revision of EDMS 19-400

7. TECHNICAL SPECIFICATION FOR TRANSFORMER COMPARTMENT

It should be furnished for all ratings of the transformer with the following:

1) The inner area of the transformer cabinet should be suitable for the dimension of all manufactures of transformers, and have additional suitable space around a transformer to facilitate the operation of airflow around the transformer, and for making the required maintenance.

2) The necessary insulated connections, bushings, etc. for medium, and low voltages sides.

3) Mechanical strength is enough to withstand lifting the whole kiosk including the transformer.

4) Completely tight from the bottom that it does not allow any oil to leak out of the compartment

5) The transformer should be earthed by connecting two copper shielded wires (25 mm2) from the two earthing terminals at the bottom of the transformer one to the main copper earthing bar of the medium voltage compartment, and the other to the main earthing bar of the low voltage compartment.

6) The single-core cables connecting the transformer to the fused switch in the medium voltage compartment should pass through an insulating Bakelite plate with 3 holes to avoid the effect of eddy currents established by single-core passing through metallic walls. The same precaution should be made for low tension bus – bars from the transformer to the L.V distribution board.

7) To avoid the scratching occur in the cable termination tubes a rubber gasket must be insulated the cable termination, and the opening of the Bakelite plate.

8) The transformer compartment should be designed to enclose 3 phase oil or dry type transformer.

9) In the case of an oil transformer (≥500 KVA), the protection should be through two levels (Alarm - Trip) of the Buchholz relay as follows:
   - The alarm will be able to open C.B (supplied with 220 V after Main C.B)
   - The trip will be able to open Tr. LBS in case (supplied with 220 V before Main C.B)

10) In case of dry transformer temperature controller will work as the following
   - The alarm will be able to open C.B when coil temperature reaches 100Cº (supplied with 220 V after Main C.B)
   - The trip will be able to open Tr. LBS in case of the coil temperature reaches 140Cº (supplied with 220 V before Main C.B)

11) In both cases, Alarm & Trip should have an indication at L.V side via flag for each case (lamps not accepted).

8. TECHNICAL SPECIFICATIONS FOR L.V. COMPARTMENT

8.1. BUS-BARS:(CU-ETP-R250 or Higher)

1) Should be according to Table-1 at latest revision of EDMS 02-300
2) All the bus-bars should be made from rectangular high-grade electrolytic copper with a purity of at least 99.99%. (its conductivity not less than 57MS/m).
3) All the bus-bars should be insulated with "heat-shrinkable" colored in red, yellow, blue, and black for the 3-phases, and neutral bus-bars, respectively.
4) The cross-section area of the neutral bus-bar should be at least 60% of the cross-section area of the phases bus-bars.
5) The cross-section area of the earth bus-bar should be at least 25% of the cross-section area of the phases bus-bars.
6) The bus-bars should be mounted on insulators made from porcelain ,cast resin or any type tested material according to the latest relative IEC standard.
7) The bus-bars, as well as, the insulators should withstand thermal, and mechanical effects resulting from short circuit current according to IEC 61439-1.

8.2. CIRCUIT BREAKERS

A. FOR INCOMINGS

- The panel includes one main 3-poles CB (ACB or MCCB) complete with an overload current, and short circuit current Protection, and without under voltage release coil to avoid voltage drop (as shown in Table-1 at latest revision of EDMS 02-300).

B. FOR OUTGOINGS C.B / SWITCH FUSE

- The panel should include several outgoing MCCB as an option/Switch Fuse (as shown in Table-1 at latest revision of EDMS 02-300), and according to the attached specifications.

C. L.V. FIXED CAPACITORS WITH M.C.C.B

1) The capacitor unit should be a dry type with self-healing properties.
2) The thickness of the dielectric insulation depends on the voltage rating. The metal part, and the capacitor's terminals make the capacitor withstand high currents, and make the capacitor's capacity value stable when operating at high operating temperatures.
3) The capacitor has the following technical data: -
   - Rated voltage (Un): 525 Volt
   - Operating voltage: 400 Volt
   - Frequency: 50 Hz
4) Discharge resistors: the capacitor discharges to 50 Volt within 60 Sec or less from an initial peak voltage of √2 times rated voltage Un.

5) Losses: Dielectric losses < 0.2 W/KVAR, total losses < 0.5 W/KVAR.
6) Maximum permissible continuous current at rated voltage, and rated frequency not less than 1.3 In.
7) Maximum inrush current at rated voltage, and rated frequency: not less than 200 In.
8) Capacitance tolerance: -5% to 10%.
9) Ambient temperature: -5 to 55°C
10) Max humidity: 95 %
11) Lifetime of capacitor at continuous normal operating conditions: not less than 130000 hours (class D).
12) Capacitor unit should have overpressure protection for 3- phase, however at vertical or horizontal position.
13) Connection: Each capacitor unit is in delta connection.
14) Insulation level: -
    - Min. Rated power-frequency. test voltage (r.m.s.): 2.5 kV
    - Rated lightning impulse voltage: 8 kV for 1.2/50 µsec.
    - Voltage withstand test.
    - Capacitance measurement.
    - Loss angle measurement on similar capacitors.

9. DESIGN CRITERIA OF L.V. COMPARTMENT

1) The inverse time-current characteristic of the release coils should comply with the latest applicable IEC standards.
2) Anywhere within the panel, the clearance distance between the main insulated bus-bars should be at least 2 cm for panels for transformers less than 500 KVA, and 4 cm for panels for transformers from 500 KVA, and above with using the supporting insulator (BB sandwich).
3) All C.B. should be adjacent, and in vertical position, the distance between the C.Bs., and between any metal part should not be less than 6 cm, and distance between switch fuse should not less than 2 cm.
4) All MCCB circuit breakers should be supplied with 3 extension insulated copper bars (6 bars as an option) of a vertical length not more than 7 cm with current density not more than 1.5A/mm2, and the middle bar must be at a different level than other adjacent bars and it shall be insulated with heat-shrinkable tube using supporting insulators when needed.
5) All the conducting parts within the panel should be made from high quality electrolytic copper purity 99 .99 % while the insulating materials should be non-flammable, and non-hygroscopic.
6) The switch fuse should come with copper droppers (with CSA 250 mm2) which are insulated with heat-shrinkable tubes to connect the cable to protect the switch fuse from being damaged due to overheating.
7) The metal parts connecting every fuse base with cable terminal should be provided with a galvanized

bolt (12 mm or 0.5 inches) with two washers, one spring washer, and one nut, all washers, and nuts should be galvanized.

8) The main contacts for fuse-carrier should be silver plated with minimum thickness of 4 microns.

9) Neutral bus bars should be prepared with holes to install feeders, and also with necessary galvanized bolts, nuts, washers, and spring washers.

10) All metal parts of the panel should have earthing bolts.

11) All cable terminals should be on a proper height (not less than 55 cm from the panel bottom), and provided with necessary clamps with a diameter not less than 8 cm to hold cables (only for panels with switch fuse)

12) The distribution panel should be designed to provide that the maximum reliability, and safety during operation, inspection, and maintenance. All the components of the distribution panel should be accessible in an easy manner.

13) All openings of incoming, and outgoing cables should be closed by a suitable method.

14) The incoming panel should be equipped with the following or digital multimeter:
- Three digital Ammeters, with scale range (0-X) A.
- One digital Voltmeter, with scale range (0-500) V.
- Three Current Transformers, class 0.5, with scale range (X/5) A.
- 3 indication Lamps (LED Type),
- Space for KWH meter with its terminals & data collector.

15) The making capacity (in KA) of all circuit breakers, and bus-bars should be at least double that of the AC current at 400V. The values should be according to IEC 60947-2.

16) The internal wiring should be non-flammable, and made of copper conductors insulated with PVC, and with a CSA of at least 2 mm2, and approved by the Egyptian electricity holding company (EEHC)

17) The insulation (PVC) of the internal wiring should withstand voltage up to 450/750 V.

18) The panel should be provided with socket 220 volt with M.C.B., and lightening lamp LED (not less than 9 watts) with M.C.B.

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10. INSPECTION, AND TESTING:

- Electricity Distribution Company (…EDC) reserves the right to witness the routine tests of the CSS.
- Unless otherwise specified or approved in writing by …EDC, all kiosks should be tested at the supplier factory in accordance with the latest applicable issues of IEC standards.
- Attendance of ...EDC representative during fabrication (assembly) stages, and testing should not relieve the supplier of his full responsibility for furnishing the kiosks in accordance of this specification requirements, and should not give him the right to invalidate any claim which …EDC, may make because of defective or unsatisfactory material or faulty workmanship.
- All tests should be carried out at the supplier`s lap, and on his expense.
- The type test certificates should be submitted with offer to...EDC.
- The tenderer should submit a quality control procedure, and approved letter with his offer.
- In addition to the type tests conducted on individual components (Transformer, Ring Main Unit and Low voltage Switchboard), CSS as a whole unit shall be type tested in accordance with the requirements of IEC 62271-202 ed.2 and the copy of type test reports should be provided and issued from reputed independent laboratories.

10.1 CSS type tests:

The CSS units shall be type tested in accordance with IEC standards 62271-202 and the following type tests have been performed and available if required:

- Tests to verify the insulation levels of CSS (IEC 62271-202/6.2)
- Tests to prove the temperature rise of main components in CSS (IEC 62271-202/6.5)
- Test to prove the capability of main and earthing circuits to be subjected to the rated peak and the rated short-time with stand currents (IEC 62271-202/6.6)
- Tests to verify the degree of protection (IEC 62271-202/6.7)
- Test to verify the withstand of the enclosure of the CSS against mechanical stress (IEC 62271-202/6.101)
- For CSS substation class (IAC-AB @ 20kA-1 sec) tests to assess the effects of arcing due to an internal fault (IEC 62271-202/6.102)
    - Tests to verify the sound level of the CSS (IEC 62271-202/Annex-BB)

10.2. CSS routine tests

The routine tests shall be made on each complete prefabricated substation or on each transport unit and, whenever practicable, at the manufacturer's factory to ensure that the product is in accordance with the equipment on which the type test has been carried out.
Each component shall be previously routine tested in accordance with their component standard, and for the assembly the following routine tests and verifications apply:

- Dielectric test on MV interconnection (IEC 62271-202/7.101)
- Test on auxiliary and control circuits (IEC 62271-202/7.102)
- Functional tests (IEC 62271-202/7.103)
- Verification of correct wiring (IEC 62271-202/7.104)

11. Marking:

- All the components of the kiosk should be marked with marking labels according to the applicable IEC standards
- Each CSS unit shall be provided with durable and clearly legible name plate which at least contains the following details:
  - Manufacturer's name
  - Type/model designation
  - Type of transformer (dry / oil)
  - KVA rating of CSS
  - Internal arc designation (If there)
  - Serial number
  - Month & Year of Manufacture
  - Tender & Purchase Order number

12. Submittals

- All necessary drawings including layout, dimensions, single line diagram of primary circuits the protection, control, and measuring circuits should be submitted with the offer.
- Technical catalogs containing the technical data for all offered components of the kiosks should be submitted
- Also, a chart illustrating selective tripping up to the before mentioned selectivity limits be submitted
- Type and special test reports/certificates from an independent testing laboratory shall be submitted to __EDC.
- Approval certificate issued by Egyptian Electricity Holding Company.

13.GUARANTEED CHARACTERISTICS

The tenderer should fill-in schedules of the technical particulars, and performance data. by signing the tender, the tenderer guarantees that the equipment supplied will confirm to technical, and performance data show in such schedules.

The supplier should guarantee the R.M.U against all defects arising out of faulty designed or workmanship or of defective materials for a period of 12 months from putting in service, or 18 months after delivery, whichever expires earlier. The good quality and manufacturing of the materials should be guaranteed. Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

14. SPARE PARTS (optional)

The tender should quote prices for the schedule of spare parts listed in the bill of quantities, and prices. Also, he may recommend some other spares which he considers necessary over a period of five years …EDC, should have the options of ordering such spare parts as they determine necessary.

- Maintenance tools-each transformer kiosk unit has to be equipped with one set of operation, and maintenance h, and tools. The list of the tools is to be included with the tender.
- The price of such tools should be included with the price for spare parts

١٥. اشتراطات :

١) يلزم تقديم شهادة الاختبار النوعى من جهه معتمدة (محايدة) وبالاضافة إلى إعتماد الشركة القابضة لكهرباء مصر.

٢) تزود كل من لوحة الجهد المتوسط ولوحة الجهد المنخفض بحامل حديد مركب عليه قفـــزان (كل قفيز به مسمارين كاملين بالصواميل والورد) لتثبيت كل من كابلات الجهد المتوسط والمنخفض أو أى وسيلة أخرى تؤدى نفس الغرض وتكون القفزان الخاصة بكابلات الجهد المنخفض فى جهه واحدة وليست متبادلة وذات قطر مناسب لكابلات الجهد المنخفض (حسب متطلبات شركة التوزيع)

٣) يلزم تركيب جوانات مطاطية لجميع الأبواب الداخلية والخارجية للكشك.

٤) قى حالة المحولات الزيتية يتم تركيب مروحة شفط (مزودة بثرموستات طبقا للمواصفات المرفقة) بغرفة المحول جهة غرفة مهمات الجهد المنخفض فى مكان يسهل التعامل معه عند فتح الباب مع الإبقاء علي باقي فتحات التهوية لضمان دورة سريان الهواء وعلى أن يتم تركيب عدد(١) شفاط للأكشاك حتى سعة ٥٠٠ك.ف.ا وعدد (٢) شفاط للسعات من ٥٠٠ ك.ف.أ فاكبر.

٥) فى حالة المحولات الجافة يتم تركيب عدد (٢) مروحة إحداهما لدفع الهواء والأخرى للشفط مزودة بثرموستات طبقا للمواصفات المرفقة بالإضافة إلى فتحات التهوية العلوية بالغرفة مع الإبقاء علي باقي فتحات التهوية لضمان دورة سريان الهواء (الشفاط من أعلى وفى الجهه الأخرى المروحة من أسفل).

٦) في حالة المحولات الجافة يلزم تركيب (controller) بلوحة مستقلة ويتم تثبيتها بمكان مناسب بلوحة الجهد المنخفض.

٧) وسيلة عدم غلق الأبواب الخارجية أثناء الصيانة تكون بواسطة خوصة مستطيلة المقطع مثبتة فى جسم اللوحة بمسمار من أحد طرفيها وبها ثقب بالطرف الأخر يتم تثبيته ببنز ملحوم بضلفة الباب الخارجى عند فتحه. .

٨) يتم تركيب عدد ٢ ترباس (وسيلة غلق معتمدة) أحدهما علوى واآلخر سفلى بالضلفة اليسرى بجميع الأبواب الخارجية لسهولة غلق وفتح جميع الأبواب .

٩) يتم تركيب عدد (٢) طفاية حريق طبقا للمواصفات المرفقة على جانبى غرفة المحول(احداهما لغرفة المحول ولوحة الجهد المتوسط والاخرى لغرفة المحول ولوحة الجهد المنخفض) مع مرور الخرطوم الحساس بكل من غرفة لوحة الجهد المتوسط وغرفة لوحة الجهد المنخفض بشكل حلقى .

١٠) جميع أبواب الكشك الأمامية و الجانبية مزودة بمفصلات من النوع ستانلس بعدد لا يقل عن ٣ مفصله / لكل دلفه باب.

١١) يجب أن تكون قاعدة أماكن دخول الكابلات من جزئيين أحدهما ثابت و الأخر متحرك (منزلق) لإحكام الغلق علي الكابلات سواء ناحية الجهد المتوسط أو المنخفض .

١٢) سقف الكشك يتمتع بخاصية سهولة الفك و إعادة التركيب طبقا لما جاء بالمواصفة بحيث يمكن للفنيين تركيب المحول من أعلى الكشك عن طريق الونش .

١٣) المسافة البينية بين قواطع الخروج للوحة الجهد المنخفض لا تقل عن ٦ سم ولا تقل عن ٢ سم في حالة استخدام سويتش فيوز للخروجات (وذلك طبقا لعدد الخروجات المذكور بمواصفة لوحات الجهد المنخفض EDMS 02- 300 جدول رقم (١))

١٤) يجب ان تكون المادة المصنوع منها الكشك الفيبر ليس لها اشعاعات ضارة وغير سامة للبيئة ولاتؤثر على الموجات الكهرومغناطيسية أو الاتصالات

١٥) يتم الفحص الظاهرى والاختبارات الروتينية للكشك عند الاستلام بمعرفة مهندسى شركات التوزيع .

١٦) التيار المقنن للقاطع عند درجة حرارة مرجعية ٥٠ درجة مئوية

١٧) أن تشتمل الملحقات على البارات (المناولات) النحاسية لتثبيت الكابلات على القاطع (بعدد ١ بارة منحنية للخارج لكل عدد ٢ بارة مستقيمة) .

١٨) يلزم مد فترة الضمان للمهمات الموردة لأول مرة الى ٢٤ شهر من تاريخ التركيب وعدد ٣٦ شهر من تاريخ التوريد.

١٩) تجهز قاعدة الكشك بكمر حديد مناسب لوزن الكشك (كامل بالمهمات) وتسمح بالتحميل الأمن للكشك من اعلى او من اسفل بحيث تضمن اتزان الكشك ويتم دهانه بطبقتين احدهما برايمر مقاوم لتأكل الحديد والأخرى اسود مع تجهيز الكمر لرفع الكشك (يمكن جلفنة الكمر طبقا لمتطلبات شركات توزيع الكهرباء).

٢٠) يجب عمل سقف مستقل لكل من لوحتي الجهد المتوسط والجهد المنخفض بالإضافة الى سقف الكشك

٢١) يلزم عمل مرايا من الصاج تغطى الكابلات اسفل السويتش فيوز وبالنسبة للقواطع لا يظهر إلا يد تشغيل القاطع فقط

٢٢) يلزم مرور كابل ربط المحول ولوحة الجهد المتوسط وكذلك بارات ربط المحول ولوحة الجهد المتوسط من خلال لوح من الفيبر العازل بسمك ال يقل عن ٤ مم مع احكام غلق الفتحات للحفاظ على درجة الحماية (IP) .

٢٣) جميع الفتحات الغير مستخدمة بين غرفة المحول وغرفتي الجهد المتوسط والجهد المنخفض

٢٤) يلزم عمل إنترلوك ميكانيكى يمنع فتح الباب إلا بعد توصيل سكينة الأرضى و يمنع إغلاق السكينة و الباب مفتوح (بند إختيارى حسب متطلبات شركات التوزيع)

٢٥) يجب أن تكون جميع الأسلاك الخاصة بالتوصيلات داخل جراب حرارى عازل.

٢٦) الأبواب الجانبية لغرفة المحول ذات دلف على مفصلات و ليست ربط بالمسامير و لها قفل داخلي بالأوكره والمفتاح.

٢٧) تزود الأكشاك بوسيلة لعدم غلق الأبواب أثناء الصيانة.

٢٨) سقف الكشك يتمتع بخاصية سهولة الفك و إعادة التركيب عن طريق عدد (٤) جوايط مسمار بالصامولة مثبتة بمكان واضح بالكشك بحيث يمكن للفنيين تركيب المحول من أعلى الكشك عن طريق الونش

٢٩) يتم تركيب المكثف الثابت خارج الكشك بحيث يركب المكثف داخل لوحة صاج منفصلة من نفس سمك صاج الكشك على الجزء العلوى للكشك على الجانب الأيمن للوحة الجهد المنخفض او يتم تركيب المكثف الثابت داخل الكشك (وذلك طبقا لطلب شركة التوزيع)

٣٠) دخول كابلات المكثف الى لوحة الجهد المنخفض يلزم أن يكون من خلال جلاند على ان تكون درجة الحماية لا تقل عن IP 54، و يلزم تأرض المكثف.

٣١) يلزم ان يكون القاطع الرئيسي للجهد المنخفض Form 2B

٣٢) يلزم لصق تعليمات تشغيل السكينة (LBS)على الباب وكذلك اتجاه فتح غلق السكينة

٣٣) بالنسبة لسكينة المحول يلزم لصق بالإضافة الى تعليمات التشغيل (لا تقم بإعادة غلق السكينة الا بعد التأكد من الراية الخاصة بفصل المحول بغرفة الجهد المنخفض فاذا كانت الراية تشير الى عطل يلزم فحص المحول والتأكد من سلامته اولا قبل التوصيل)

٣٤) يلزم لصق تعليمات على مراية القاطع الرئيسي (اذ وجدت القاطع في وضع trip لا تعيد توصيل القاطع قبل التحقق من وضع الراية الخاصة بفصل القاطع) .

١٦. <u>المواصفـات الفنيـة لمراوح شفط الـهواء لحجـرات المحـولات :</u>

1. جهد التشغيل للموتور 220 فولت + 10% 50 ذبذبة / ثانية وجه واحد.
2. قدرة الموتور لا تزيد عن 180 وات عند معامل قدرة.
3. تصريف المروحة لا يقل عن 3800 م3 / ساعة تحت ضغط جوي ، ويتم تقديم شهادة اختبار تفيد ذلك.
4. السرعة لا تزيد عن 1400 لفة / دقيقة
5. المروحة من النوع الذى يتحمل العمل بصفة مستمرة (24) ساعة بدون توقف.
6. ريش المروحة من الصاج المجلفن أو مكسو بالمعدن أو الألومنيوم – بحيث تكون الريش لها قدرة تحمل عالية للتشغيل المستمر والحرارة العالية وبقطر 40 سم كاملة الإتزان داخل شبكة من السلك

7. الظروف الجوية لعمل الموتور والمروحة تحت درجة الحرارة للهواء المسحوب من الحجرة تصل الى 65م°.
8. تعمل المروحة من خلال ثرموستات (مرفق المواصفات الفنية للثرموستات) يتم تثبيتها بغرفة المحول جهة الجهد المنخفض بجوار الباب.
9. درجة عزل الموتور class F ودرجة الحماية IP 54 OR IP 55
10. تكون المروحة من النوع قليل الضوضاء.
11. يجب اعتماد مواصفات المروحة من شركتنا قبل التوريد.

17. SPECIFICATIONS OF THE TRANSFORMER COMPARTMENT AIR TEMPERATURE CONTROLLER THROUGH CONTROLLING OPERATION OF TRANSFORMER COMPARTMENT VENTILATING FAN.

THE REQUIRED IS: A thermostat of any suitable design complete with all accessories (and a relay or contractor if the switching rating of the thermostat is low than the specified herein rating), and a suitable prepared controller housing, in order to verify the following requirements, and specifications:

1- The purpose of the controller is to open the electrical circuit of the transformer compartment ventilating fan, (which is installed in a window highly positioned in one of the transformer Compartment walls) when the ventilating air temperature drops down to a certain adjustable value (between 30˚C, 80˚C), and closes this electric circuit when room air temperature rises 5˚C-8˚C above that adjust value.

2- System (fan supply) voltage: 220 (+10% volts, -15% volts)

3- Thermostat specifications:

- Temperature range: 30˚C - 80˚C
- Max working temperature at thermostat heat: 80˚C
- Temperature differential (the difference between the opening, and closing temperature of the thermostat): 5˚C - 8˚C
- Manufacturing tolerance in opening, and closing temperature: low as possible.
- Combined controller output contact rating (for fan operation):
    - Resistive: > 16 A
    - Inductive (full load ampere): > 5 A at p.f. 0.6 or 0.4
- The controller housing should be prepared for wall mounting.

١٨. المواصفات الفنية لتوريد وتركيب نظام إطفاء تلقائى بإستخدام الخرطوم الحساس بجهاز إطفاء ثانى أكسيد الكربون

الإستخدام :

يتم إستخدام ذلك النظام المعتمد على خراطيم الإستشعار الحساسة وذلك لتغطية الأماكن الضيقة والتى يصعب الوصول إليها

مكونات النظام :

١. جهاز إطفاء بغاز ثانى أكسيد الكربون سعة 10 كجم.
٢. وحدة رأس جهاز الإطفاء مزودة بمانوميتر.
٣. خرطوم الإستشعار.
٤. طبه مناسبه للخرطوم.
٥. وسيلة مناسبه ومحكمه لتعليق بدن جهاز الإطفاء

جهاز الإطفاء :

١. يكون بدن الجهاز من الصلب المسحوب على البارد ومطابقا للمواصفات القياسية المصرية (م.ق.م 780)وأن يتحمل ضغط التفجير ≤400 بار وضغط اختبار هيدروليكى ≤200 بار.
٢. يكون التصنيع طبقاً للمواصفات القياسية المصرية 735 لسنة 2006
٣. يجب أن يحتوى البدن على وسيلة لتعليق الجهاز بواسطة حامل يكفل له التثبيت بطريقة مأمونة إلى الحائط أو إلى الأرض
٤. لا يحتوى البدن على فتحات بخالف فتحة مجموعة الرأس فقط ألا يقل قطرها الداخلى عن 19 مم
٥. ألا يقل سمك البدن عند القمة والقاع عن 6مم .
٦. أن يكون البدن مدهون من الخارج بدهان الكتروستاتيك باألحمر الالمع ، ومعالج من الداخل ضد الصدأ (فسفته)
٧. أن يحقق التفريغ للمحتويات 90% بدون تجميد.
٨. يجب أن يعالج بدن الجهاز بالوسائل المناسبة والكفيلة بإزالة أية زيت أو شحوم أو مواد عازلة قبل الطلاء وبما يوفر درجة عاليه من مقاومة الصدأ والتماسك بطبقة الطلاء والتى يجب أن يتوفر فيها الجودة العالية ومقاومة العوامل الجوية والخدوش والصدمات.
٩. أن يكون الجهاز قد اجتاز جميع االختبارات المنصوص عليها في المواصفة القياسية المصرية (735/2006)
١٠. أن يوضح على الجهاز البيانات الأتية بشكل واضح يصعب محوه :
   - إسم الصانع وعلامته التجارية وعنوانه والسجل التجاري والصناعي.
   - جهاز ثاني أكسيد الكربون سعة (--) كجم وتشمل كمية الغاز األسميه ونوع الملأ.
   - طريقة إستعمال الجهاز (مختصرة وواضحة)
   - عبارة (تعاد تعبئة الجهاز بعد الإستعمال مباشرة)
   - تاريخ صنع الجهاز و الرقم المسلسل للجهاز.
   - رقم المواصفة المصنع عليها الجهاز و وزن الجهاز فارغاً بمشتمالته ووزن الجهاز معبأ.

خرطوم الإستشعار :

1. خرطوم مصنوع من مـادة بوليمارية عـالية الجـودة.
2. ذات مرونـة عـالية.
3. حـساس لدرجة الحرارة وينفجر الخرطوم عند أعلى نقطة تأثراً من 80 100: درجة.
4. عند الإنفجار تتشكل فتحة الخرطوم كـأنها مخرج لنشر المـاده الخـاصـه بالإطفاء.
5. لا يتـفاعل مع المواد الكيميائية.
6. أن يكون القطـر الخـارجى للخرطوم 6 مم والقطـر الداخلى 4 مم.
7. ضغط التشغيل 20 بار.
8. وحدة رأس جهاز الإطفاء
9. تصمم مجموعة الرأس من النحاس المسبوك تحت ضغط والمطلى بطبقه من النيكل كروم.
10. أن تكون الرأس معتمده طبقا للمواصفات الأوروبية .
11. تصمم مجموعة الرأس بحيث تشكل آلية جيدة للتحكم في التشغيل الإيقاف وتكون مأمونة
12. مزوده بصمام أمان لتصريف الشحنة الزائدة من الغاز عند ضغط يتراوح من 183 إلي 200 كجم/سم

شروط عامة لنظام الإطفاء:

1. أن يتم التنفيذ وفقاً لأصول الصـناعة.
2. أن يتم تثبيت خرطوم الإستشعار بطريقه فنية مناسبة
3. يجب تقديم شهادة ضمان ضد عيوب الصناعة
4. يجب تقديم شهادة من الهيئة المصرية العامة للمواصفات والجودة (التوحيد القياسي) أو شهادة موثقة من المصنع الرئيسي عند الإستلام
5. يجب تقديم شهادة منشأ أو شهادة من هيئة الرقابة على الصادرات والواردات عند الإستلام.

19.GUARANTEE TABLES
Guarantee Table (1) For ENCLOSURE & Bus bars

1- Enclosure:

- Manufacturer's name :-----------
- Type :-----------
- Material :----------
- Thickness before painting :---------- mm
- Thickness of coating :---------- mm
- Dimensions :---------- mm

Width * Depth * Hight : -------- mm

2-Bus-bars: -

- Material : ---------
- Material purity : ---------
- Material conductivity--MS/m
- Cross section area

  * Per phase--mm^2
  * Neutral bus --- mm^2
  * Earth bus--mm^2

- Minimum clearance between phases --------------------------------------- mm
- Minimum clearance to earth--mm
- Continuous current inside the enclosure

* At 50 ˚C for indoor-- A

- Dynamic withstand current, peak-- KA
- One- second thermal withstand current--------------------------------------- KA
- Rated insulation voltage--V
- One- Minuit power frequency withstand voltage--------------------------- V
- Heat-shrinkable tube insulation 12/20 Kv------------------------------kV
- Leakage path of supporting insulators ----------------------------------- kV

3-Heaters: -

- Rated Power : -------- W
- Hygrostat :--------- Yes/No
- M.C.B :--------- Yes/No

4-No. of E.F.I :---------

- Rated operating current :--------- A
- Rated operating voltage :--------- V

I / We guarantee the technical data given above for the offered equipment.

Signature:.......................................

Date: ....

GUARANTEE TABLE No. 2

L.V Main Circuit – Breakers

Manufacturer's name:	-----------
Standard specification:	-----------
Type:	-----------
Rated current inside the enclosure at 50 ˚C:	-----------**A**
Rated insulation voltage:	-----------**V**
Rated breaking capacity at 400 V:	-----------**kA**
Rated making capacity, peak:	-----------**kA**
Rated current of thermal release inside the enclosure (Ir) At 50 ˚C:	-----------**A**
Thermal release setting value at 50 °C:	-----------**A**
Short circuit release setting value:	-----------**A**
One-second power frequency withstand voltage:	-----------**V**
Material of main contacts:	-----------
Material of arcing contacts:	-----------
Type of arc control device;	-----------
Material of arc control device:	-----------

I / We guarantee the technical data given above for the offered circuit –breakers.

Signature:.........................

Date: ....

GUARANTEE TABLE No. 3

Outgoing Circuit – Breakers (Option)

Manufacturer's name:	----------
Standard specification:	----------
Type:	----------
Rated current inside the enclosure at 50 °C:	----------**A**
Rated insulation voltage:	----------**V**
Rated breaking capacity at 400 V:	----------**kA**
Rated making capacity, peak:	----------**kA**
Rated current of thermal release inside the enclosure (Ir) At 50 °C:	----------**A**
Thermal release setting value at 50 °C:	----------**A**
Short circuit release setting value:	----------**A**
One-second power frequency withstand voltage:	----------**V**
Material of main contacts:	----------
Material of arcing contacts:	----------
Type of arc control device;	----------
Material of arc control device:	----------

I / We guarantee the technical data given above for the offered circuit –breakers.

Signature:............

Date: ....

EEHC

GUARANTEE TABLE No. 4
Outgoing Switch fuse (Option)
Fuse-links of rated current 400 A.
For …. KVA transformer panel

PARTICULARS

1. Manufacturer's name	
2. Standard specifications applied	
3. Rated current	------Amp.
4. Rated voltage	------volts
5. Switching voltage limits	------volts
6. Frequency	------ Hz
7. Rated power loss	------W
8. Time /Current characteristic attached with the offer	Yes / No
9. Pre-arcing Time /Current characteristic attached with the offer	Yes / No
10. Rated breaking capacity	------KA
11. Cut - off characteristics, attached with the offer	Yes / No
12. I^2 t characteristics, attached with the offer	Yes / No
13. Material, and cross-section area of main contacts	------mm^2
14. Surface contact area of main contacts with holders at each end	------mm^2
15. Material of coating of copper contacts	
16. Material, and cross-section of fusing metal	------mm^2
17. Material of insulating part	
18. Weight of fuse	------gm
19. Dimensions of fuse:	
a) Length	------mm
b) Width	------mm
c) Height	------mm

I / We guarantee the technical data given above for the offered Fuse links.

Signature:..

Date: ....

EEHC

GUARANTEE TABLE No. 5
Outgoing Switch fuse
(Option)Fuse –base for fuses
630 A.
For …. KVA transformer panel

PARTICULARS

1- Manufacturer's name	
2- Standard specifications applied	
3- Rated Current	------Amp
4- Rated voltage	------Volts
5- Frequency	------Hz
6- Rated breaking capacity	------kA
7- Material, and cross-section area of main contacts (width × thickness)	------mm
8- Surface contact area of main contact with fuse link	------mm2
9- Material of coating of copper contacts	------
10- Material of insulating part	------
11- Weight of fuse base	------gm
12- Dimensions of fuse base:	
a) Length	------mm
b) Width	------mm
c) Height	------mm

I / We guarantee the technical data given above for the offered Fuse bases.

Signature: ……………………………….

Date: ……………………………….

EEHC

GUARANTEE TABLE No. 6
FOR
CABLE LOAD-BREAK SWITCHES

Item:	**Specs.**
Type, and manufacturer	
Nominal operating voltage	---------kV
Maximum operating voltage	---------kV
Nominal current at 40°c	---------A
Leakage path of supporting insulators	---------mm/kV
Thermal withs and current for one sec (r.m.s)	---------kA
Making current (peak)	---------kA
Minimum clearance in air:	
• between phases	---------mm
• between phase, and earth	---------mm
• pole center	---------mm
Transformer off-load breaking capacity (PF=0.15)	---------A
Capacitive breaking current	---------A
▪ Technical specifications for the Earthing switch:	
- Making current (peak)	---------
- Mechanical endurance class	---------
- Thermal withs and current for one sec (r.m.s)	---------

I / We guarantee the technical data given above for the offered Load-break switches.

Signature:........................

Date: ....

GUARANTEE TABLE
No. 7FOR
TRANSFORMER LOAD-BREAK SWITCHES

- Rated voltage ------kV
- Rated load-breaking current (P.F.O.7) ------A
- Basic impulse level ------kV
- Power frequency withstand voltage ------kV
- Leakage path of supporting insulators ------kV/cm
- Minimum clearance in air:
    - Between poles ------mm
    - To earth ------mm
    - Pole center ------mm
- Overall dimensions: Length x Width x Height cm × cm × cm
- Rated impulse current I_{dyn} ------kA
- Rated short-time current I_{th} 1sec ------kA
- Rated making current I ma ------kA
- Transformer off-load breaking capacity (P.F = 0.15) ------A
- Capacitive breaking current (P.F = 0.15) ------A
- Number of switching operations at:
    - Rated breaking current (P.F =0.7). ------

I / We guarantee the technical data given above for the offered Load-break switches.

Signature:.............................

Date: ....

GUARANTEE TABLE
No. 8FOR
GENERAL DATA OF LOAD-BREAK SWITCH

Item	Specif.
▪ The contact resistance per phase	----------
▪ Opening time for load break switch	----------
▪ Minimum Mechanical endurance Operating cycle M1	YES/NO
▪ Minimum Electrical endurance class E3	YES/NO
▪ Capacitive current breaking switch class C2	YES/NO
▪ The thickness of galvanizing coating for all steel parts should be not less than 12µm.	YES/NO
▪ The terminals of connection should be silver-plated with thickness not less than 5µm.	YES/NO

I / We guarantee the technical data given above for the offered Load-break switches.

Signature: ....
Date: ....

EEHC